Rational Functions: Full Practice Exam: Parent Key

Worked solutions and proposed practice rubric. Keep separate from the student copy.

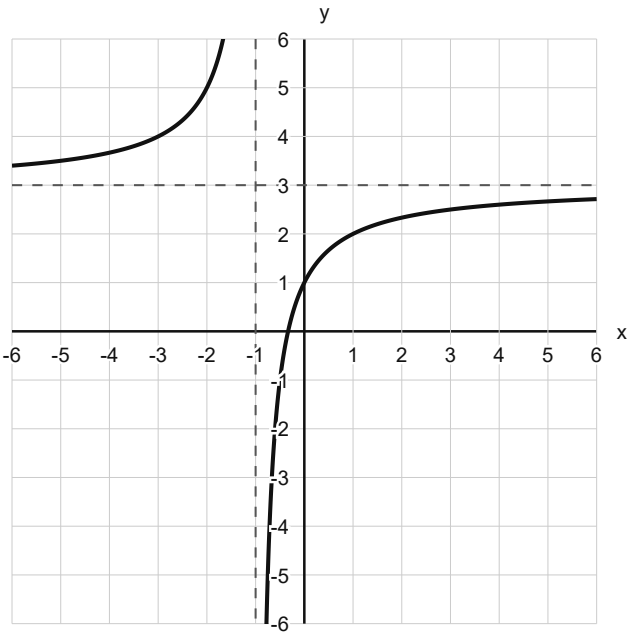
13 questions | 58 points | 60 minutes (user-selected allowance)

Award follow-through credit; do not repeatedly deduct an originating error. Accept equivalent valid functions and their own correct hole heights. This is not an official school key.

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1. 8 points

1. The denominator vanishes at $x = -1$. The reciprocal term tends to zero at either end, so the horizontal asymptote is $y = 3$.
2. For example, $f(-3) = 4$, $f(0) = 1$, and $f(1) = 2$. Left of the vertical asymptote the curve lies above 3; right of it the curve lies below 3.
3. As $x \to -\infty$, $f(x) \to 3$; as $x \to +\infty$, $f(x) \to 3$. It approaches from above on the left and below on the right, but the stated limit in each direction is sufficient for the end-behavior points.



Practice rubric (8 points)

- 4: Graph: correct asymptotes 1, at least two correct labeled points 1, consistent branches 2.
- 2: Far-left limit 3.
- 2: Far-right limit 3.

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2. 4 points

1. Factor $x^2 - 4 = (x - 2)(x + 2)$. The original denominators exclude -2 and 2.
2. The numerator over the shared denominator is $3(x - 2) - (2x - 1) = x - 5$. The result is $\dfrac{x - 5}{(x - 2)(x + 2)}$, with both original exclusions.

Practice rubric (4 points)

- 1: Valid common denominator.
- 2: Correct subtraction and simplified numerator.
- 1: Both original excluded inputs.

3. 4 points

1. Factor the product as $\dfrac{(x - 5)(x + 3)}{(x - 5)(x + 5)} \cdot \dfrac{x + 5}{(x - 3)(x + 3)}$.
2. Cancellation gives $\dfrac{1}{x - 3}$, but the original denominator zeros $-5, -3, 3, 5$ remain excluded.

Practice rubric (4 points)

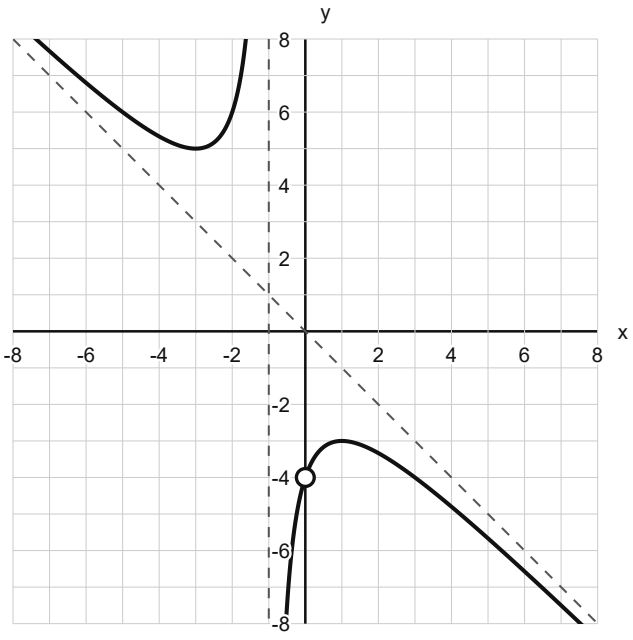
- 1: Correct factorizations.
- 2: Valid cancellation and final expression.
- 1: All four original exclusions.

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4. 8 points

1. A rule used to draw this graph is $F(x) = -x - \dfrac{4}{x+1}$, with $x \neq -1, 0$. This rule is parent-only; students read the printed graph.
2. Increasing intervals are $(-3, -1) \cup (-1, 0) \cup (0, 1)$. The open circle at 0 breaks the domain, even though the branch continues in the same increasing direction on either side.
3. The relative minimum is $(-3, 5)$; the relative maximum is $(1, -3)$. Concavity is up on $(-\infty, -1)$, down on $(-1, 0) \cup (0, \infty)$.
4. The asymptotes are $x = -1$ and $y = -x$; there is no horizontal asymptote. Accept equivalent wording that clearly excludes the hole. No differentiation is required.



Practice rubric (8 points)

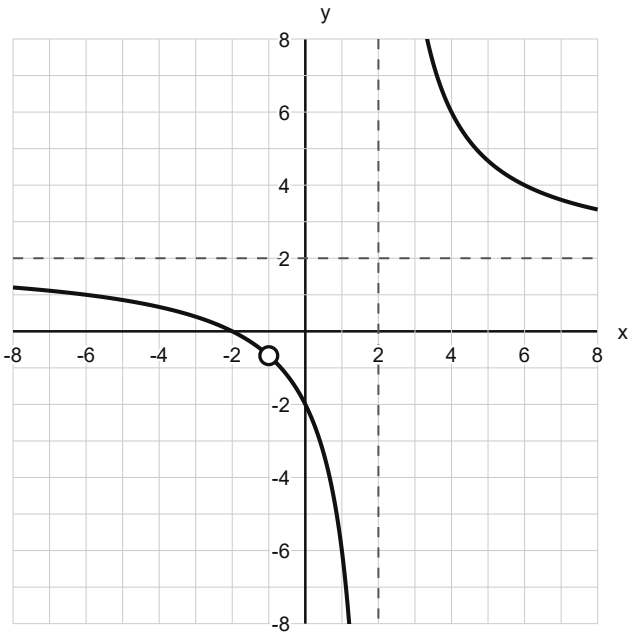
- 1: Increasing intervals with the missing input excluded; preserve partial credit for correct direction/boundaries.
- 2: Both relative extrema with types and coordinates.
- 2: Both concavity directions, with the hole excluded.
- 3: Vertical line, slant line and no-horizontal conclusion, one point each.

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5. 10 points

1. Factor $R(x) = \dfrac{2(x+1)(x+2)}{(x+1)(x-2)} = \dfrac{2(x+2)}{x-2} = 2 + \dfrac{8}{x-2}$, with original exclusions -1 and 2.
2. The intercepts are $(-2, 0)$ and $(0, -2)$. At the canceled input, the simplified rule gives $2(-1+2)/(-1-2) = -2/3$, so the hole is $(-1, -2/3)$.
3. Vertical asymptote $x = 2$; horizontal asymptote $y = 2$. There is no slant asymptote. Left of 2 the curve lies below 2 and decreases toward negative infinity near the vertical asymptote; right of 2 it lies above 2 and decreases toward the horizontal line.
4. The graph must have an open circle at the hole and must pass through the actual y-intercept, not the horizontal-asymptote height.



Practice rubric (10 points)

- 2: Factoring and valid simplification.
- 2: Both intercepts, one point each.
- 2: Hole location and height.
- 2: Vertical and horizontal asymptotes.
- 2: Consistent labeled sketch including open hole and correct branch approach; avoid double-penalizing a single copied feature error.

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6. 4 points

1. Division gives $S(x) = -\frac{x}{2} - 1 + \frac{1}{2x-4}$. Multiplying $(-x/2 - 1)(2x - 4)$ gives $-x^2 + 4$, leaving remainder 1.
2. The slant asymptote is $y = -x/2 - 1$; there is no horizontal asymptote. The remainder fraction tends to zero. As $x \to -\infty$, $S(x) \to +\infty$; as $x \to +\infty$, $S(x) \to -\infty$.

Practice rubric (4 points)

- 1.5: Correct division or equivalent exact identity.
- 1: Correct slant line and absence of horizontal line.
- 0.5: Far-left direction.
- 0.5: Far-right direction.
- 0.5: Remainder fraction tends to zero.

7. 4 points

1. Rewrite $C(n) = 12 + \frac{60}{n+10}$. For every finite $n \geq 0$, the fraction is positive.
2. As production grows, the fraction decreases toward zero. The average cost approaches $12 per kit from above, but is strictly greater than $12 for every finite number of additional kits.

Practice rubric (4 points)

- 1: Limit 12.
- 1: Correct interpretation and dollars per kit.
- 1: Approach from above/decreasing.
- 1: Nonattainment at finite n with valid justification.

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8. 4 points

1. One valid rule is $G(x) = \dfrac{3(x+4)(x-2)}{2(x+1)(x-2)}$, with $x \neq -1, 2$. Its remaining numerator has only the zero -4, its remaining denominator vanishes at -1, its leading ratio is 3/2, and the shared factor creates a hole at 2.
2. For this rule the missing height is $3(2+4)/(2(2+1)) = 3$, giving $(2, 3)$. Accept any rule meeting all four conditions and its own correctly calculated hole height; a unique rule or height is not required.

Practice rubric (4 points)

- 1: Correct only x-intercept, supported by the rule/check.
- 1: Correct only vertical asymptote, supported by the rule/check.
- 1: Correct horizontal height, supported by the rule/check.
- 1: Only hole at 2 and its height for the chosen rule: 0.5 each.

9. 2 points

1. The nonzero curve approaches the x-axis at both far ends, so its horizontal asymptote is $y = 0$. Its numerator degree must be smaller than its denominator degree.
2. Equal degrees give a nonzero leading-coefficient ratio; larger numerator degree does not give a zero horizontal asymptote. A specific pair of degrees cannot be determined from the graph alone.

Practice rubric (2 points)

- 1: Choice B.
- 1: Explanation connects zero horizontal end height to smaller numerator degree.

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10. 2 points

1. D is odd, crosses the origin, has vertical asymptotes at $x = \pm 2$, and approaches $y = 0$. For $x > 2$, its values are negative, matching the right outer branch.
2. A is even and has nonzero horizontal height; B has the opposite sign; C has no real vertical asymptotes. Accept any decisive, correct graphical distinction rather than requiring every listed feature.

Practice rubric (2 points)

- 1: Choice D.
- 1: A decisive valid graph-feature justification.

11. 2 points

1. A simplifies to $-\dfrac{(x-1)(x-4)}{2(x+2)}$ while retaining the excluded input 0. It has a hole at $(0, -1)$, not a y-intercept.
2. The zeros are 1 and 4; the vertical asymptote is -2; division gives slant $y = -x/2 + 7/2$. B fills in the missing point, C moves the vertical asymptote, and D reverses the graph vertically. The hole is the decisive difference between A and B.

Practice rubric (2 points)

- 1: Choice A.
- 1: Justification distinguishes the hole/domain from the otherwise equivalent rule B.

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12. 2 points

1. Both ends of f tend to negative infinity. For $m = 1$ or 3, the leading behavior of g is $-8x^2$, and for $m = 4$ it is $-5x^2$.
2. For $m = 5$, the leading behavior is $3x^3$, so the left end is down and right end up. That does not match both ends down. Matching end behavior means matching the two directions, not identical degree or growth rate.

Practice rubric (2 points)

- 1: Choice C.
- 1: Leading-term/parity reasoning explaining the mismatch.

13. 4 points

1. Near 3 the values approach a finite height near 4 from both sides; with the stated assumptions this supports the hole at $x = 3$. Near 1 the magnitudes grow without bound, supporting the vertical asymptote $x = 1$.
2. The roles of the two excluded inputs are reversed from the mini quiz. The rounded samples support the requested inferences, not proof of an exact limit or unique rational rule. One generating rule is $3 + 2/(x - 1)$, additionally excluding 3; students are not asked to find it.

Practice rubric (4 points)

- 1: Hole at 3.
- 1: Finite-height table evidence for the hole.
- 1: Vertical asymptote at 1.
- 1: Unbounded-magnitude table evidence for the asymptote.